

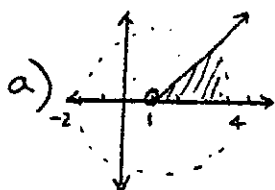
Question 1 (20 marks)**Marks**

- (a) Sketch the region represented in the complex plane by $|z-1| < 3$ and $0 \leq \text{Arg}(z-1) \leq \frac{\pi}{4}$. **3**
- (b) Use De Moivre's theorem to solve the equation $z^5 = 1$. Show that the points representing the five roots of this equation on an Argand diagram form the vertices of a regular pentagon of area $\frac{5}{2} \sin \frac{2\pi}{5}$ and perimeter $10 \sin \frac{\pi}{5}$. **5**
- (c) If $z_1 = 24 + 7i$ and $|z_2| = 6$, find the greatest and least values of $|z_1 + z_2|$. **4**
- (d) (i) Express $-1 + i\sqrt{3}$ in mod - arg form. **2**
(ii) Hence evaluate $(-1 + i\sqrt{3})^6$. **2**
- (e) Find the equation of the locus of z in the complex number plane such that $|z-5| = 2|z+1|$. Describe the locus geometrically. **4**

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Question 1

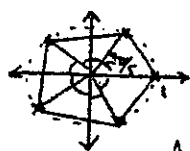


b) $z^5 = 1$

$|z| = 1$

$\arg z = \frac{2\pi k}{5}$ where $k = 0, 1, 2, 3, 4$
 $= 0, \pm \frac{2\pi}{5}, \pm \frac{4\pi}{5}$

$z = \cos(\pm \frac{2\pi}{5}), \cos(\pm \frac{4\pi}{5}), 1$

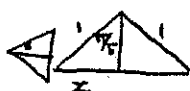


Pentagon vertices subtend
 an angle of $\frac{2\pi}{5}$ at centre.

Area of one sector $= \frac{1}{2} r^2 \sin \theta$

$\therefore \text{Area} = 5 \times \frac{1}{2} \times 1^2 \sin \frac{2\pi}{5}$

$= \frac{5}{2} \sin \frac{2\pi}{5}$



$x = \sin \frac{\pi}{5}$

$\therefore \text{Perimeter} = 10 \sin \frac{\pi}{5}$

c) $|z_1| = 25$

$|z_1 + z_2| \leq |z_1| + |z_2| = 31$

(Greatest if collinear)

Least if z_1, z_2 are collinear
 in opp. directions.

$|z_1 + z_2| \geq ||z_1| - |z_2|| = 19$

$\therefore \text{Greatest} = 31, \text{Least} = 19$

d) i, $|z| = 2$ $\arg z = \tan^{-1}(-\sqrt{3})$

$= \frac{2\pi}{3}$
 $\therefore 2 \cos \frac{2\pi}{3}$

ii) $(-1 + i\sqrt{3})^6 = 2^6 \cos(\frac{2\pi}{3} \times 6)$

$= 64 \cos 4\pi$

$= 64$

1- dotted circle

1- correct sector

1- (1,0) excluded

1- use of De Moivre's

1- correct roots.

1- diagram of pentagon

1- correct derivation of area

1- correct derivation of perimeter

1- calculates $|z_1|$

1- adds modulus to find greatest

1- subtracts to find least (± 19)

1- takes abs value to obtain 19

1- correct modulus

1- correct angle in correct quadrant

1- use of De Moivre's

1- complete simplification to 64

e) let $z = x + iy$

$$|x + iy - 5| = 2|x + iy + 1|$$

$$\sqrt{(x-5)^2 + y^2} = 2\sqrt{(x+1)^2 + y^2}$$

$$x^2 - 10x + 25 + y^2 = 4(x^2 + 2x + 1 + y^2)$$

$$x^2 - 10x + 25 + y^2 = 4x^2 + 8x + 4 + 4y^2$$

$$3x^2 + 18x + 3y^2 - 21 = 0$$

$$x^2 + 6x + y^2 - 7 = 0$$

$$(x+3)^2 + y^2 = 16$$

Circle centre $(-3, 0)$

radius 4.

1 - substitutes $z = x + iy$
or similar

1 - simplified equation.

1 - recognition of circle form

1 - centre + radius.